Sewing measures and identity limits of the recoil two-point spectrum

Alan Day Chengheng*

Working research note — 31 October 2026

Abstract

We distinguish the identity representation in real-parameter Virasoro TQFT from a normalized massless endpoint of the recoil two-point spectrum. The common cone-angle continuation used in the paired amplitude cannot reach the vacuum of both Virasoro factors. The reduced spectrum nevertheless admits a controlled delta-function limit. We determine its pole-crossing correction and a finite-mass error bound. The previously exhibited insertion freedom survives this endpoint test. These statements do not construct a de Sitter sewing inner product or a physical finite-gravity four-point function.

1 What the known sewing rule fixes

We use CEZ [1], version 3, whose HTML equations have continuous numbering. In the standard block normalization, for real $b > 0$ and positive real internal momenta, equations (23)–(25) give

$$\langle s|s'\rangle = \frac{\delta(s-s')}{\rho_s(s)}, \tag{1}$$

$$\rho_s(s) = \rho_0(s) C_0(P_1, P_2, s) C_0(P_3, P_4, s), \tag{2}$$

$$\rho_0(P) = 4\sqrt{2}\sinh(2\pi b P)\sinh(2\pi P/b). \tag{3}$$

Thus the four-punctured-sphere resolution is $\mathbf{1} = \int_0^\infty \mathrm{d}s\, \rho_s(s)|s\rangle\langle s|$. Multiplying each block by the square roots of its vertex C_0 factors changes the measure to ρ_0, as in CEZ (27)–(28). Dropping the vertices without this basis change is inconsistent.

With the convention $|s\rangle = \int_0^\infty \mathrm{d}t\, F_{st}|t\rangle$, preservation of the inner product requires

$$\int_0^\infty \mathrm{d}t\, \frac{F_{st}\overline{F_{s't}}}{\rho_t(t)} = \frac{\delta(s-s')}{\rho_s(s)}. \tag{4}$$

This follows from the displayed block metric; CEZ (55) and (122) supply the corresponding exchange and fusion-inverse relations. The bilinear inverse relation and the sesquilinear isometry must not be confused. We have not evaluated the full fusion kernel numerically in this note.

A change $|s\rangle \mapsto a_s|s\rangle$, with nonzero a_s, leaves (4) covariant if

$$\rho'_s = \rho_s/|a_s|^2, \qquad F'_{st} = a_s F_{st}/a_t. \tag{5}$$

This is basis freedom. It does not show that an arbitrary change of a physical operator insertion preserves crossing with a fixed fusion kernel.

The identity limit in CEZ (29) is distributional:

$$\lim_{P_3 \to iQ/2} C_0(P_1, P_2, P_3) = \frac{\delta(P_1 - P_2)}{\rho_0(P_1)}, \qquad Q = b + b^{-1}. \tag{6}$$

*AI research assistant (OpenAI).

The momenta $P_1, P_2 > 0$ remove reflection redundancy; the full-line convention has an additional reflected delta. The vacuum is outside the normalizable continuum. CEZ (46) distinguishes an incoming or external identity from an outgoing continuum label. In particular, squaring (6) pointwise is not a definition of a paired amplitude: δ^2 is undefined.

2 Why this is not yet the de Sitter measure

CLP [2], section 6, distinguishes analytic continuation of fixed-topology amplitudes from a de Sitter theory with arbitrary sewing. CEM [3], equations (20), (24) and section 5, uses a late-time Teichmüller/moduli-space inner product which ordinary conformal blocks do not diagonalize. These are statements about the versions inspected here, not an exhaustive claim about all subsequent literature.

For a nonorthogonal basis, the required formal resolution is instead

$$\mathbf{1} = \iint \mathrm{d}a\ \mathrm{d}b\ |a\rangle (G^{-1})_{ab} \langle b|, \qquad G_{ab} = \langle a|b\rangle. \tag{7}$$

Existence of the inverse on a suitable test space, treatment of null directions, positivity and convergence are additional tasks. Neither the analytically continued Cardy factor nor the observer weight $z(m) = m^2 e^{-2\pi m}$ has thereby been identified with the physical Plancherel measure.

3 The common-angle path cannot reach two vacua

Use exactly the two factors in CLP (3.2), (3.4), (3.6) and (3.44):

$$b_\pm^2 = \pm i/\kappa, \qquad P_\pm = \pm \frac{i\alpha}{4\pi b_\pm}, \qquad h_\pm = \frac{(b_\pm + b_\pm^{-1})^2}{4} + P_\pm^2. \tag{8}$$

For a common angle $\alpha = 2\pi(1 - \nu/\kappa)$, direct algebra gives

$$h_+ = \frac{1 - i\nu}{2} + \frac{i(1+\nu^2)}{4\kappa}, \tag{9}$$

$$h_- = \frac{1 + i\nu}{2} - \frac{i(1+\nu^2)}{4\kappa}, \qquad h_+ + h_- = 1. \tag{10}$$

This exact label identity is consistent with CLP's framing discussion. At $\nu = i$, the pair is $(1, 0)$; at $\nu = -i$, it is $(0, 1)$. At $\nu = 0$ neither factor is the vacuum. Here $h_\pm$ label the two continued factors; their sum is not being identified with the physical scalar dimension.

On the positive-angle branch, the separate vacuum conditions require

$$\alpha_\pm^{\mathrm{vac}} = 2\pi(1 + b_\pm^2) = 2\pi(1 \pm i/\kappa). \tag{11}$$

They are distinct at finite κ. The reflected negative-angle solutions do not restore a common angle. Although these angles approach each other as $\kappa \to \infty$, their difference is of the order retained by the fixed-mass scaling. It cannot be discarded when testing a vacuum limit.

Consequently, the scalar relation $m^2_{\mathrm{field}} = 1 + \nu^2$ does not justify identifying $\nu = \pm i$ with a double-vacuum leg of the paired TQFT graph. Continuing the real-ν paired expression meromorphically with a common complex ν is also different from taking a fresh modulus square at each complex ν. A double-vacuum calculation needs independently specified labels and a pairing prescription.

4 A controlled endpoint of the reduced spectrum

We now change the question explicitly. Work at radius one, after the fixed-mass $\kappa \to \infty$ limit, and continue only the established reduced two-sided spectrum

$$k_\nu(\omega) = \frac{1}{8\pi[\cosh \pi\omega + \cosh \pi\nu]}, \qquad g_\nu(0) = \frac{\nu}{4\pi \sinh \pi\nu}. \tag{12}$$

Set $\nu = i(1-\epsilon)$, $0 < \epsilon < 1$. Dividing by the divergent $g_\nu(0) = (1-\epsilon)/(4\pi \sin \pi\epsilon)$ gives

$$p_\epsilon(\omega) = \frac{\sin \pi\epsilon}{2(1-\epsilon)[\cosh \pi\omega - \cos \pi\epsilon]}, \tag{13}$$

$$\chi_\epsilon(t) = \int_{\mathbb{R}} \mathrm{d}\omega\, e^{-i\omega t} p_\epsilon(\omega) = \frac{\sinh((1-\epsilon)t)}{(1-\epsilon)\sinh t}. \tag{14}$$

At $t = 0$ the continuous value is one. No $1/(2\pi)$ occurs in this forward transform. The identity follows by analytic continuation inside $0 < \operatorname{Im} \nu < 1$, where compact subsets admit integrable spectral domination, or by residues.

The density is nonnegative and integrates to one. Differentiating (14) at zero gives

$$v_\epsilon := \int_{\mathbb{R}} \omega^2 p_\epsilon(\omega)\, \mathrm{d}\omega = \frac{2\epsilon - \epsilon^2}{3}, \qquad \int_{|\omega|>D} p_\epsilon(\omega)\, \mathrm{d}\omega \le \frac{v_\epsilon}{D^2}. \tag{15}$$

Hence $p_\epsilon \rightharpoonup \delta$. For a C^2 test function with bounded second derivative, Taylor's formula and evenness give the explicit error $\|f''\|_\infty v_\epsilon/2$. Concentration and normalization also prove weak convergence for bounded continuous tests. This is a normalized infrared endpoint, not a finite unnormalized massless-field correlator or a construction of the identity operator on a gravitational Hilbert space.

The pole crossing is visible

The poles are $\omega = i(2n \pm \epsilon)$. The nearest two, $\pm i\epsilon$, pinch the real axis at zero. Their residues are respectively $\pm[2\pi i(1-\epsilon)]^{-1}$. No pole crosses before the endpoint on the positive-ϵ path. Continuing through it is a different matter.

For $\epsilon = -\delta$, $0 < \delta < 1$, the same real-axis integral is

$$\int_{\mathbb{R}} \mathrm{d}\omega\, e^{-i\omega t} p_{-\delta}(\omega) = -\frac{\sinh((1-\delta)t)}{(1+\delta)\sinh t}. \tag{16}$$

It differs from the continued right side of (14) by

$$\chi_{-\delta}(t) - \int_{\mathbb{R}} \mathrm{d}\omega\, e^{-i\omega t} p_{-\delta}(\omega) = \frac{2\cosh(\delta t)}{1+\delta}. \tag{17}$$

This is precisely the sum of the two crossing-pole contributions; it also follows from the hyperbolic addition formula. The difference survives at $t = 0$. We have tracked this reduced two-pole problem, not all poles of the finite-κ double-sine kernel.

5 Finite observer mass does not remove the freedom

Fix $M > m_* > 0$, $D = 2(M - m_*)$, $A = \cosh(2\pi M)$. Define

$$R_{M,\epsilon}(\omega) = \left(1 - \frac{\omega^2}{4M^2}\right) \frac{A - \cosh \pi\omega}{A - \cos \pi\epsilon}, \tag{18}$$

$$q_{M,\epsilon,\lambda}(\omega) = \mathbf{1}_{[-D,D]}(\omega) p_\epsilon(\omega) R_{M,\epsilon}(\omega) \left(1 + \frac{\lambda\omega^2}{4M^2}\right), \quad \lambda \ge 0. \tag{19}$$

This is a declared spectral model, with $0 \le R \le 1$ on its support. Let $T_D = \int_{|\omega|>D} p_\epsilon$ and $v_D = \int_{-D}^{D} \omega^2 p_\epsilon$. The exact mass loss is

$$1 - \int q_{M,\epsilon,0} = T_D + \frac{v_D}{4M^2} + \frac{\sin \pi\epsilon}{(1-\epsilon)(A - \cos \pi\epsilon)} \left(D - \frac{D^3}{12M^2} \right). \tag{20}$$

To prove it, expand $1 - R$. In its second term, $\cosh \pi\omega - \cos \pi\epsilon$ cancels the spectral denominator, leaving a quadratic polynomial to integrate. Using (15), set

$$B_{M,\epsilon} = \frac{v_\epsilon}{D^2} + \frac{v_\epsilon}{4M^2} + \frac{\sin \pi\epsilon}{(1-\epsilon)(A - \cos \pi\epsilon)} \left(D - \frac{D^3}{12M^2} \right) = O_{M,m_*}(\epsilon). \tag{21}$$

For all real t, the triangle inequality gives

$$|\widehat{q}_{M,\epsilon,\lambda}(t) - \chi_\epsilon(t)| \le B_{M,\epsilon} + \frac{\lambda v_\epsilon}{4M^2}. \tag{22}$$

The same bound controls $\|q_{M,\epsilon,\lambda} - p_\epsilon\|_1$. Therefore every fixed $\lambda \ge 0$ yields the same weak limit $q \rightharpoonup \delta$; restoring the common factor gives $z(M)\delta$. Yet these same choices change the first finite-mass correction on the real principal series. Leading QFT matching plus this normalized endpoint test thus fails to fix the insertion.

More generally, a fixed bounded continuous even factor, or a family converging uniformly to one on this finite interval, is tested only through its limiting diagonal value $n(M, 0)$. This is a necessary normalization condition in the chosen spectral convention, not a proof that all such factors extend to a crossing-consistent physical four-point function.

6 Limits, verification and the next physical task

The identity limit is not uniform over time:

$$\lim_{\epsilon \downarrow 0} \chi_\epsilon(t) = 1 \quad (t \text{ fixed}), \qquad \lim_{|t| \to \infty} \chi_\epsilon(t) = 0 \quad (\epsilon > 0), \tag{23}$$

and for fixed $\tau \neq 0$,

$$\lim_{\epsilon \downarrow 0} \chi_\epsilon(\tau/\epsilon) = e^{-|\tau|}. \tag{24}$$

Bound (22) transfers the latter result to q at fixed M, m_*, λ. This is an infrared model timescale, not a finite-gravity chaos exponent.

The saved diagnostics contain 106 passed checks: 16 real-axis Fourier/moment checks, 18 crossing checks, 36 finite-mass identity, bound and precision comparisons, and 36 label-algebra checks. The last group includes repeated vacuum checks across different probe-label loops; it is not 36 independent physical tests. Calculations used 50 decimal digits and selected finite-mass integrals were repeated at 65 digits; the largest change was 6.41×10^{-52}. This is an empirical refinement, not interval certification. Parameters, versions, outputs and hashes are stored with the code. Large-time scaling entries evaluate the exact transform; they are not claimed as oscillatory quadratures.

The next concrete task is the global $\mathrm{SL}(2, \mathbb{C})$ partial-wave inner product in the $G = 0$ limit, including reflection and spectral measure, followed by an explicit comparison with the regularized recoil graph H. A Barnes-integral resemblance is insufficient. Any double-vacuum study must instead keep independent labels and track the continuation of already-sewn expressions. Physical finite-κ sewing, finite-G four-point dynamics and a complete de Sitter dual remain open.

References

[1] S. Collier, L. Eberhardt and M. Zhang, *Solving 3d Gravity with Virasoro TQFT*, arXiv:2304.13650v3.

[2] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1.

[3] S. Collier, L. Eberhardt and B. Mühlmann, *A microscopic realization of* dS_3, arXiv:2501.01486v1.